

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (IT) Examination

November 2012

MA206 Probability, Statistics & Numerical Analysis

Date: 05.11.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Find median and mode of the following data. [04]

(i) 8, 12, 20, 40, 8, 7, 5

(ii) 7, 7, 8, 6, 5, 8, 3, 2, 2, 9, 7.

(b) The correlation coefficient between two variables x and y is 0.48 and the co-variance between them is 36. If the variance of x is 16, find S.D. of y . [03]

Q - 2 (a) Calculate Karl Pearson's coefficient of correlation between expenditure on advertising and sales from the data given below. Give your comment on r . [06]

Advertising expenditure ('000 Rs.) : 39 65 62 90 82 75 25 98 36 78

Sales (lakh Rs.) : 47 53 58 86 62 68 60 91 51 84

(b) Find the Median for the distribution of the weights of 150 students from the data given below: [04]

Weight in kg.:	30-39	40-49	50-59	60-69	70-79	80-89
Frequency:	18	37	45	27	15	8

(c) The weights of 54 college students are given in the following table. Find the Quartile Deviation of the distribution. [04]

Weights (Kg)	60 - 64	65 - 69	70 - 74	75 - 79	80 - 84	85 - 89
No of Students	5	9	16	12	8	4

OR

Q - 2 (a) An industrial engineer collected the following data on experience and performance rating of 8 operators. [06]

Operators	1	2	3	4	5	6	7	8
Experience (year)	16	12	18	4	3	10	5	12
Performance rating	87	88	89	68	58	80	70	85

Estimate the performance rating of an engineer who has 9 and 15 years of experience?

(b) Obtain mean and standard deviation for the following data. [04]

Age in year	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of Members	3	61	132	153	140	51	2

- (c) A psychologist wanted to compare two methods A and B of teaching. He selected a random sample of 22 students. He grouped them into 11 pairs so that the students in a pair have approximately equal scores on an intelligence test. In each pair the student was taught by method A and the other by method B and examined after the course. The marks obtained by them are tabulated below. [04]

Method A	24	29	19	14	30	19	27	30	20	28	11
Method B	37	35	16	26	23	27	19	20	16	11	21

Find the rank correlation coefficient.

- Q - 3 (a) If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, $P(A \cap B) = \frac{1}{6}$, Find $P(A \cup B)$, $P(A' \cup B')$, and $P(A' | B')$ [06]
- (b) The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = \frac{1}{2}$. What is the probability that a repair time exceeds 2 hours? [04]
- (c) The mean and standard deviation of the wages of 6000 workers engaged in a factory are Rs. 1200 and Rs. 400 respectively. Assuming the distribution to be normal estimate: [04]
- Percentage of workers getting wages above Rs. 1600
 - Number of workers getting wages between Rs. 600 and Rs. 900.
 - Number of workers getting wages between Rs. 1100 and 1500.

OR

- Q - 3 (a) Write the area properties of Normal distribution, also draw the respective graphs [06]
- (b) Suppose that after 10 years of service, 30% of computers have problems with Motherboards (MB), 40% have problems with hard drives (HD), and 15% have problems with both MB and HD. What is the probability that a 10-year old computer still has fully functioning MB and HD? [04]
- (c) It is known that the probability that an item produced by a certain machine will be defective is 0.1. If a sample of ten items is selected at random from the output of the machine, determine the probability that a sample will contain Exactly two defective items. Use Binomial distribution. [04]

SECTION - II

- Q - 4 (a) Define (i) Absolute Error (ii) Relative Error and (iii) Percentage Error. Three approximate values of the number $\frac{1}{3}$ are given as 0.30, 0.33 and 0.34. Which of these three is the best approximation? [04]
- (b) Use Bisection method to determine a real root of the equation $x^3 - x - 1 = 0$ in the interval $[1, 2]$. Perform three iterations only [03]
- Q - 5 (a) Define (i) Forward difference (ii) Backward difference. Construct the forward difference table for the function $y = 2x^3 - 2x^2 + 3x - 1$ for $x_0 = 1, x_1 = 2, x_2 = 3$. [06]

- (b) Use Newton's formula of Interpolation and determine the cubic polynomial which takes the following value: [04]

$$x: 0 \quad 1 \quad 2 \quad 3$$

$$y: 1 \quad 0 \quad 1 \quad 10$$

Find y when $x = 0.5$

- (c) Certain experimental values of x and y are given below: [04]

$$x: 0 \quad 2 \quad 5 \quad 7$$

$$y: -1 \quad 5 \quad 12 \quad 20$$

If $y = a_0 + a_1x$, find approximate values of a_0 and a_1 by method of least squares

OR

- Q - 5 (a) Construct the Lagrange interpolation polynomial from following table values: [04]

$$x: 0 \quad \frac{1}{6} \quad \frac{1}{2}$$

$$y = \sin \pi x: 0 \quad \frac{1}{2} \quad 1$$

Find $\sin \pi \frac{1}{4}$

- (b) The following table give the value of $\log_{10} x$ [04]

$$x: 300 \quad 304 \quad 305 \quad 307$$

$$\log_{10} x: 2.4771 \quad 2.4829 \quad 2.4843 \quad 2.4871$$

Use Newton's divided difference formula and find $\log_{10} 301$

- (c) Determine the constants a and b by method of least squares such that $y = ae^{bx}$ fits the following data: [06]

$$x: 2 \quad 4 \quad 6 \quad 8 \quad 10$$

$$y: 4.077 \quad 11.084 \quad 30.128 \quad 81.897 \quad 222.62$$

- Q - 6 (a) Evaluate numerically $\int_0^1 \frac{1}{1+x} dx$ by trapezoidal rule with $h = 0.125$ [06]

- (b) Given $\frac{dy}{dx} = y - x$ where $y(0) = 2$ Find $y(0.1)$ and $y(0.2)$ correct to four decimal places using Runge-Kutta fourth Order formula. [08]

OR

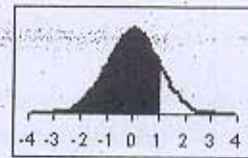
- Q - 6 (a) Evaluate numerically $\int_0^1 \sqrt{1+2x} dx$ using the Gaussian formula with $n = 4$ ordinates. [07]

Use table values

n	i	u_i (ordinate)	W_i (Weight)
4	1; 4	∓ 0.86113631	0.34785484
	2; 3	∓ 0.33998104	0.65214516

- (b) Determine the value of y when $x = 0.1$, using modified Euler's method. Given that $y(0) = 1$, and $\frac{dy}{dx} = x^2 + y$. Take $h = 0.05$ [07]

Date: 05/11/2012



Distribution Function of Standard Normal Random Variable

591

Table 3 Distribution Function of Standard Normal Random Variable

z	0	1	2	3	4	5	6	7	8	9
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7703	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9278	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9430	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9648	.9656	.9664	.9671	.9678	.9686	.9693	.9700	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9762	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9874	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.	.9987	.9990	.9993	.9995	.9997	.9998	.9998	.9999	.9999	1.0000

Note 1: If a normal variable X is not "standard," its value must be standardized: $Z = (X - \mu)/\sigma$. Then $F_X(x) = F_Z\left(\frac{x - \mu}{\sigma}\right)$.

Note 2: For $z \geq 4$, use $F_Z(z) = 1$ to four decimal places; for $z \leq -4$, $F_Z(z) = 0$ to four decimal places.

Note 3: The entries opposite $z = 3$ are for 3.0, 3.1, 3.2, etc.

Note 4: For $z < 0$ use $F_Z(z) = 1 - F_Z(-z)$.

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